MILITARY SPECIFICATION

RESISTOR, FIXED, WIRE-WOUND (POWER TYPE),

ESTABLISHED RELIABILITY,

STYLE RWR89

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for style RWR89, established reliability, power type, wire-wound, fixed resistor. These resistors are available with terminal types "S" and "W" (solderable or weldable, inductively wound) and terminal "N" (solderable, noninductively wound), and terminal "Z" (weldable, noninductively wound).

1.2 Military part number. Resistors covered by this specification shall be identified by a military part number which shall consist of the style designation, terminal and winding, coded resistance value, tolerance, and failure rate level. The military part number shall be derived in accordance with MIL-R-39007 and shall be in the following form:

RWR89	S	1000	F	S
Style	Terminal and winding	Coded resistance	Resistance tolerance	Failure rate level

2. APPLICABLE DOCUMENTS

2.1 Government document.

2.1.1 Specification. The following specification forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of this document shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-R-39007 - Resistors, Fixed, Wire-Wound (Power Type), Established Reliability, General Specification for.

(Copies of the specification required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-39007, and as specified herein.

3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3.2.1 Noninductive winding. Noninductively wound resistors are identified by terminal designation "N" to indicate a solderable terminal, and "Z" to indicate a weldable terminal, noninductively wound resistor.

* 3.3 Minimum resistance value for "S" and "W" (inductively wound) type resistors. For inductively wound resistors, the minimum resistance value be 10 ohms for all tolerances.

* 3.3.1 Minimum resistance value for "N" and "Z" (noninductively wound) type resistors. For noninductively wound resistors, the minimum resistance value shall be 0.1 ohm. For resistance tolerance B (.1 percent) minimum resistance value shall be .499 ohm.

* 3.4 Maximum resistance value for "S" and "W" type resistors. The maximum resistance value based on use of 0.001 inch (0.03 mm) nominal diameter wire (with a nominal resistivity of 300 ohms per circular mil foot) shall be 3,570 ohms. The maximum resistance value based on use of 0.0008 inch (0.020 mm) nominal diameter wire (0.00072 inch (0.0183 mm) absolute diameter) (not for NASA use) shall be 4,120 ohms. Resistors below 950 ohms in value shall be wound with 0.00175 inch (0.044 mm) or larger diameter wire.

* 3.4.1 Maximum resistance value for "N" and "Z" type resistors. For resistors having noninductive characteristic "N" and "Z" designations, the maximum resistance value shall be 1,780 ohms for 0.001 inch (0.03 mm) nominal diameter wire. For 0.0009 inch (0.020 mm) nominal diameter wire (not for NASA use), the maximum resistance value shall be 2,050 ohms.

3.5 Power rating. The power rating shall be 3 watts at 25°C.

3.6 Weight. The maximum weight shall be .00662 pound (3 grams); for "N" type resistors, the maximum weight shall be .00882 pound (4 grams).

* 3.7 Terminal lead types. This style resistor is available in both solderable (S) and (N) and weldable (W) and (Z) leads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-39007.

4.2 Short-time overload. The duration of applied voltage shall be 5 seconds. Voltage applied shall be equivalent to 5 times rated wattage.

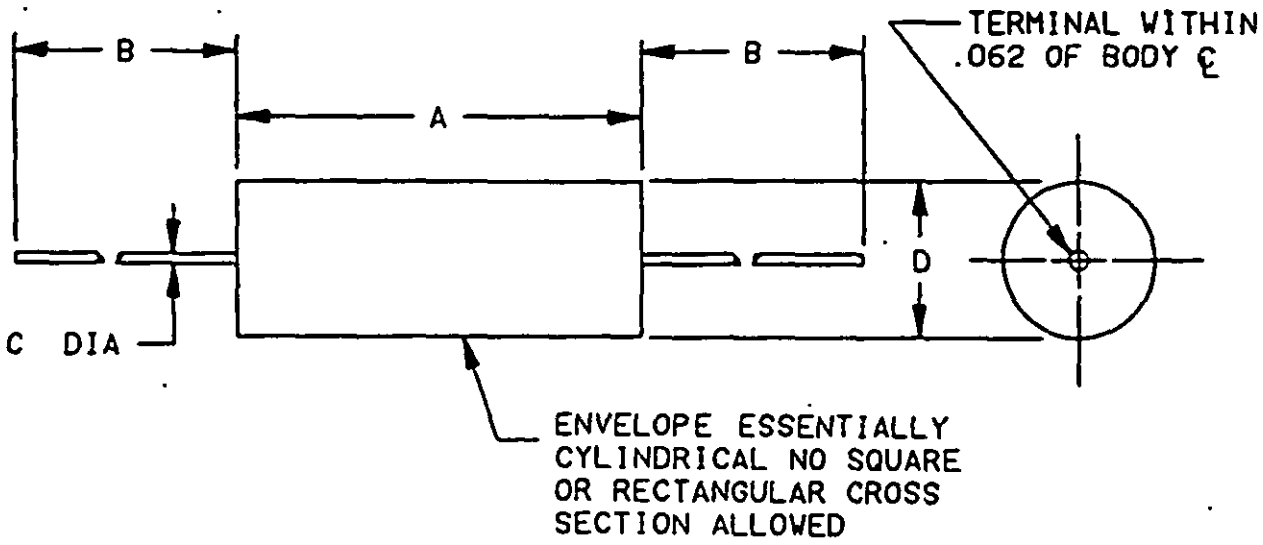
4.3 Dielectric withstanding voltage.

4.3.1 Atmospheric pressure. The magnitude of test voltage shall be 1,000 volts rms.

4.3.2 Barometric pressure (reduced). The test voltage shall be 200 volts rms.

4.4 Terminal strength. The applied force shall be 10 pounds.

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Standard style	Dimensions			
	A±.062	B,min	C±.002	D±.031
RWR89	.560	1.500	.032	.187

Inches	mm
.002	0.05
.031	0.79
.032	0.81
.062	1.57
.187	4.75
.560	14.22
1.500	38.10

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension "A" is "clean lead to clean lead".
4. Lead length for tape and reel packaging shall be 1 inch minimum.

FIGURE 1. Style RWR89 resistor.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39007.

6. NOTES

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-R-39007 are applicable to this specification.

6.2 MIL-R-26 substitution data. Resistors of this specification regardless of the failure rate designation are substitutes for resistors of the same resistance value and tolerance specified in the inactive specification MIL-R-26/5, style RW79. Also, resistors of this specification are substitutes for new design for specification MIL-R-26/4, style RW69. However, they are not directly interchangeable.

6.2.1 MIL-R-39007/3 substitute data. For replacement purposes use style RWR89 of MIL-R-39007/11 over the deleted RWR69 of MIL-R-39007/3.

6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-39007/11F	Resistor, Fixed, Wire-Wound (Power Type), Established Reliability, Style RWR89

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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